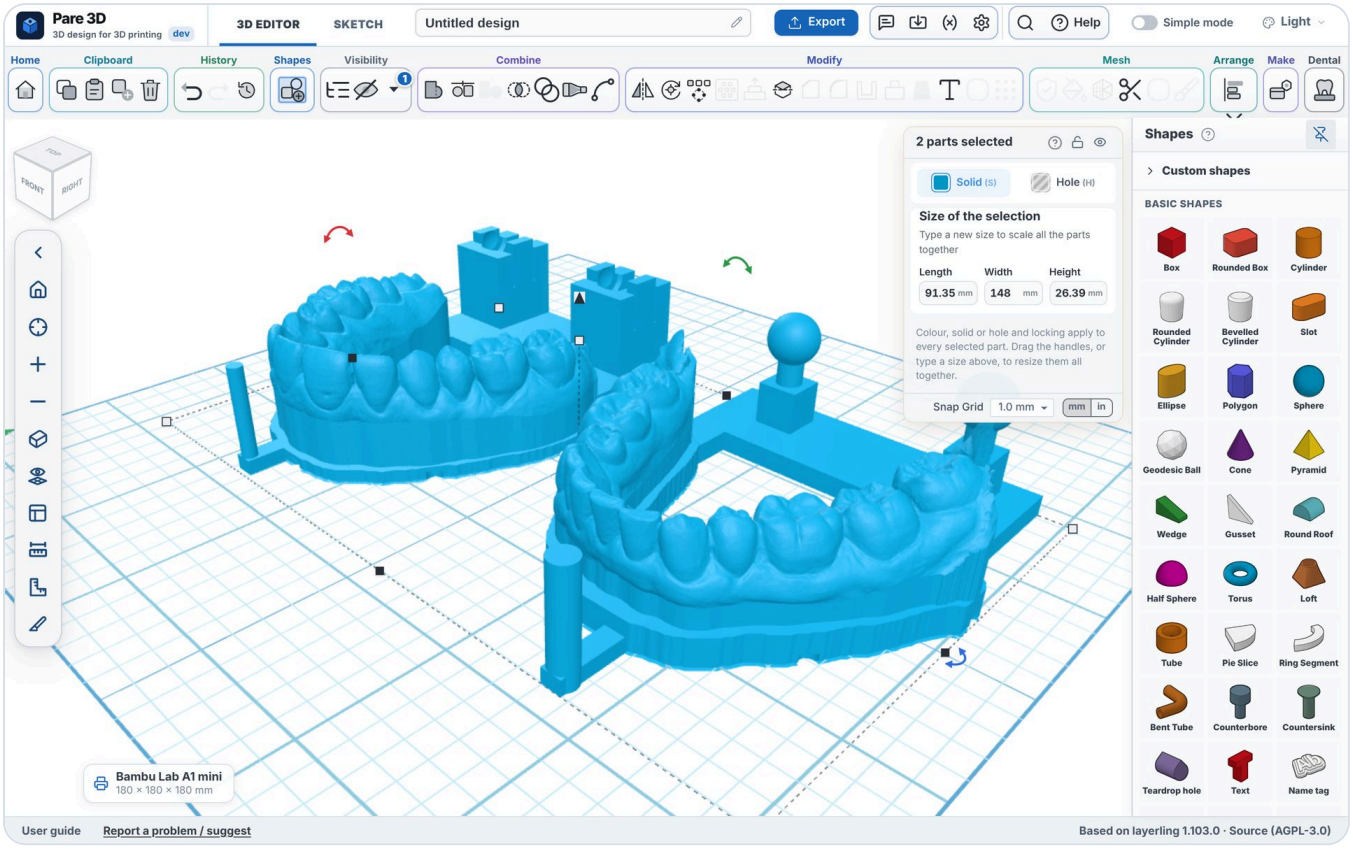
Pare 3D · Dental Studio · Build 77 · October 2026

# The Model Builder Guide

From intraoral scans to printed study, ortho, aligner, crown-and-bridge and articulated models - six steps, every setting explained for clinicians.




An upper and lower pair made from three intraoral scan files, with a snap-in ball articulator behind the molars and an incisal pin, laid out ready to print.

Screens shown are from Pare 3D build 77 on a 1440 × 900 window. Scans used: a full upper arch (330,577 triangles), a full lower arch and a buccal bite scan from the author's own mouth.

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**The short way**

Home page › **Dental models from scans** › drag the upper, lower and bite files onto the window › pick what the model is for › tick **Print as an articulator** if you want one › **Make models** › **Send to** your slicer. Everything else in this guide is for when you want to change what Pare 3D chose.

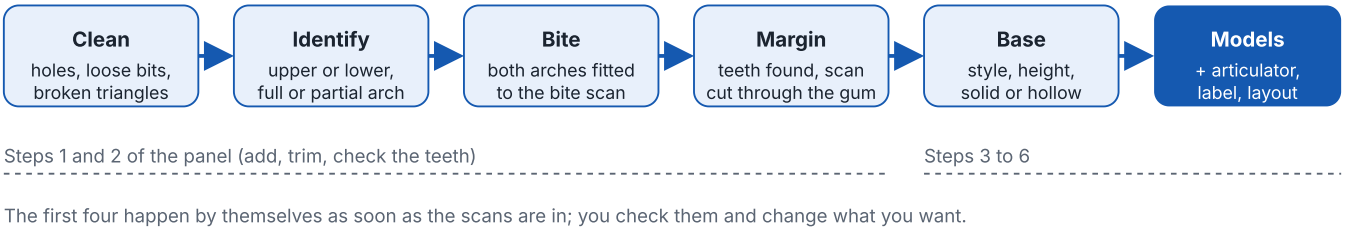
**New in builds 76 and 77**

This guide can be downloaded from Dental Studio itself (the download button in its title bar). A quadrant against a full arch is articulated as a full pair, on a horseshoe base (chapter 9). *Trim* is now *Trim and fill*: beside Delete, **Erase & Fill** closes a bubble, spike or void with a smooth patch; the warning counts only the crowns of the teeth (within 6 mm of the cusp tips), not the gum, palate or flaps; **Undo** and **Redo** buttons step back through each delete or fill; importing a full-arch scan offers to raise the triangle limit for cutting and grouping in one click; and the gum cut leaves fewer thin slivers along the margin. Build 75 brought the six step cards, the side-view articulator picture and the corrected height prediction.

# 1 What Dental Studio does

Dental Studio turns raw intraoral scans - open surfaces that end where the scanner stopped - into closed, printable models with a base, and, for a pair, matching bases in the bite and an optional articulator.

It does the work a dental lab technician or a model-builder program (Medit Model Builder, 3Shape Model Builder) does, in one panel and in this order:




The first four happen by themselves as soon as the scans are in; you check them and change what you want.

## What is never changed

- **The tooth surface.** The scanned teeth and the gum beside them are kept exactly as scanned. They are only turned and moved as a whole; no point of the kept surface moves relative to another. Trimming and margin cuts never cross a tooth.
- **The bite.** When a bite scan lines the arches up, both arches are moved by one rigid transform each, so the occlusion is that of the bite scan - and that bite is locked in: trimming, sculpting or cutting the palate away never changes it (chapter 5).

## What you get

- One model per arch, closed and printable, named after the scan (*<scan name> upper model*, *<scan name> lower model*), with a base of the kind you chose.
- For a pair: matching bases that stand in the bite, laid out side by side on the build plate or stacked in occlusion, and - if you ask - an articulator built into the bases.
- A notice saying what was made (heights, hollow or solid, wall thickness) and any warnings.

**Where it runs**

Everything happens on your own computer, in the Pare 3D app or browser tab; scans are never uploaded. An articulated pair from full-arch scans took about 50 s on a small 2-core test machine; a recent desktop is faster. The window stays responsive while it works.

# 2 Before you start: scans and files

Dental Studio needs the arch scans as surface meshes, in millimetres, exported from the same case so they share the scanner's coordinates.

## File types

| Format | Use | Notes |
|---|---|---|
| STL | Most common export from all scanners | Geometry only. Pare 3D reads binary and text STL. |
| PLY | Medit, 3Shape and others export colour this way | Read as geometry; scan colour is not used for the model. |
| OBJ | Some scanners and labs | Geometry is used; textures are ignored. |
| 3MF | Exported from other programs | Each mesh in it comes in as a part. |

## What to export from the scanner

1. **Upper arch** - the maxillary scan.
2. **Lower arch** - the mandibular scan.
3. **Bite scan** (optional but recommended for a pair) - the buccal bite registration: the scan of the teeth in occlusion from the buccal side. Most scanners export it as a third file, or include it in the case export.

> **Export all three from the same case, without moving them**
> Scanners export the arches and the bite in one shared coordinate system, usually already in occlusion. Dental Studio starts from those positions. A scan that was moved, mirrored or re-oriented in another program before export may not line up; Pare 3D then says so (*The two scans are not where the scanner put them...*) and makes each arch on its own.

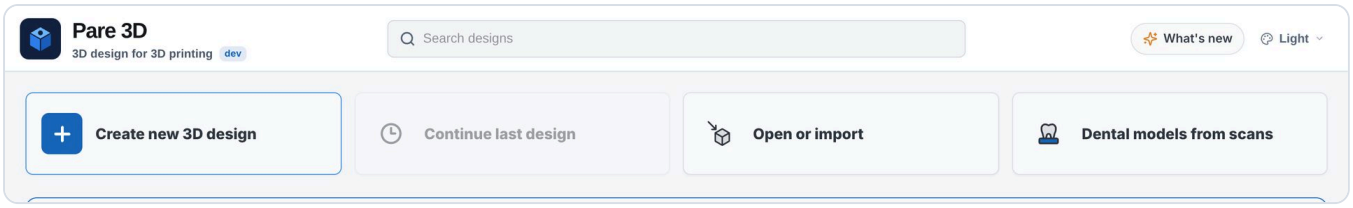
## Scan quality that matters

- **Captured depth.** Scan far enough into the vestibule and lingual sulcus for the base you want. The panel reports how many millimetres of scan lie below the occlusal plane for each arch; an ortho base needs the most.
- **Holes and loose islands.** Small holes are filled and loose fragments are removed automatically. Large gaps on the teeth themselves cannot be invented - rescan those areas.
- **Partial arches** (a quadrant or sextant) are recognised as partial and get a block base; the tie bar option is for full arches only (chapter 4, *Full arch or quadrant*).
- **Units.** Files must be in millimetres. A scan that comes in a few millimetres or several metres across was exported in another unit.

> **Patient files**
> Name files with a patient code rather than the patient's name if the designs or exports will be shared. The Label option (chapter 10) engraves a code on the model.

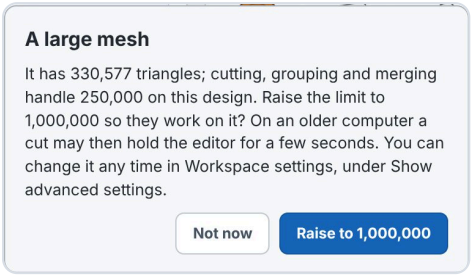
# 3 Opening Dental Studio

There are three ways in; all end in the same panel.




The home page. **Dental models from scans** opens a new design with Dental Studio already open; **What's new** lists what each build brought.

1. **From the home page** - click **Dental models from scans**. A new, empty design opens with Dental Studio ready (fastest for a new case).
2. **By dropping the files** - drag the scan files from your desktop or a folder onto the editor window. A dashed frame shows while you hold them over it; they come in when you let go. Pare 3D recognises intraoral scans and offers **Dental Studio** in a notice at the bottom of the window.
3. **From the toolbar** - the tooth button in the **Dental** group, or search `Ctrl` + `K` for *dental*, *articulator* or *model builder*. With scans selected, those are used; with none selected, every scan in the design is offered.

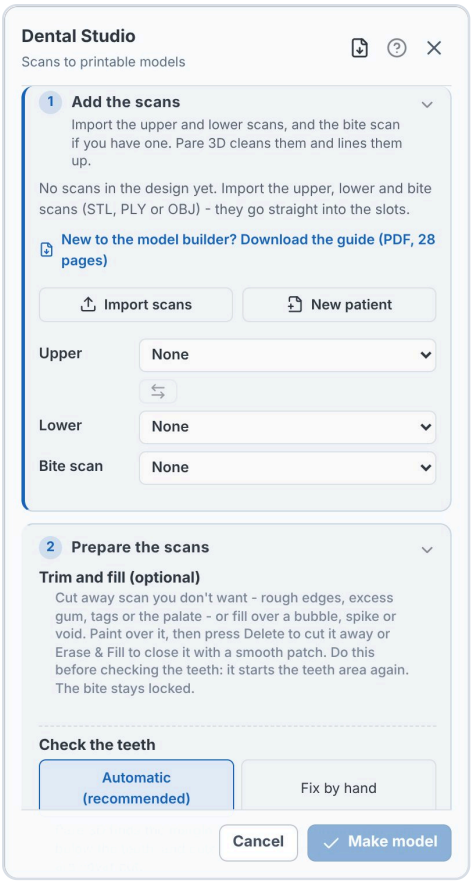



Offered once a session when a scan has more triangles than cutting and grouping handle.

## A large mesh: raise the limit once

Cutting, grouping and merging an imported mesh are worked out in the browser, up to a limit of 250,000 triangles by default. A full-arch scan has more (330,577 for the upper used here). When one comes in, a card offers to raise the limit for the design to room for twice the mesh - at least 1,000,000. Press **Raise to 1,000,000**: Dental Studio itself does not need it, but grouping a model with a label or cutting it with a hole does. On an older computer a cut can then hold the editor for a few seconds.

The card asks once a session. The setting stays in **Workspace settings** under **Show advanced settings** (*Cut imported meshes up to*), to change at any time.



Dental Studio before any scan is in.

## The panel: six step cards

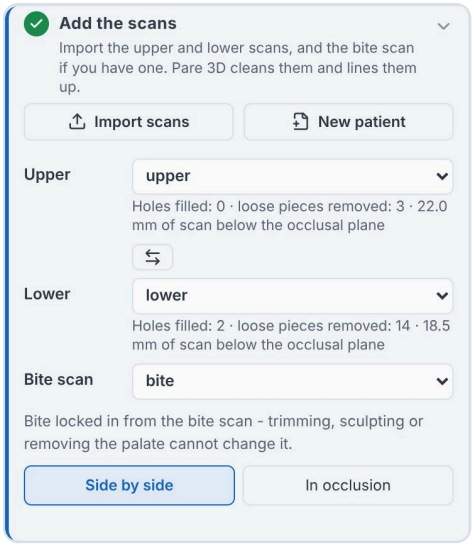
Each step sits on a card of its own, top to bottom. A step's number turns into a green tick once it is complete and you have worked in it. Click a step's title to fold it to one line that sums up its settings, and again to open it; the card you last worked in is marked with a blue edge. The line above the buttons says what is still needed - for example *All set - press Make models*.

1. **Add the scans** - which scan is upper, lower and bite; side by side or in occlusion.
2. **Prepare the scans** - trim or fill what you do not want, then check the teeth.
3. **Choose the model** - what it is for, and solid or hollow.
4. **Base** - its shape, details and the model's height.
5. **Articulator** - only shown for a pair.
6. **Name the model** - a label, or no patient details.

With a single arch the articulator card is left out, so there are five. The panel can be dragged by its title bar; the download button beside **?** saves this guide as a PDF (also linked in step 1 before any scan is in); **?** opens this part of the user guide; **×** or **Cancel** closes it without changing anything.

# 4 Step 1 - Add the scans

Bring the files in, check that each landed in the right slot, and read the cleaning report under each arch.



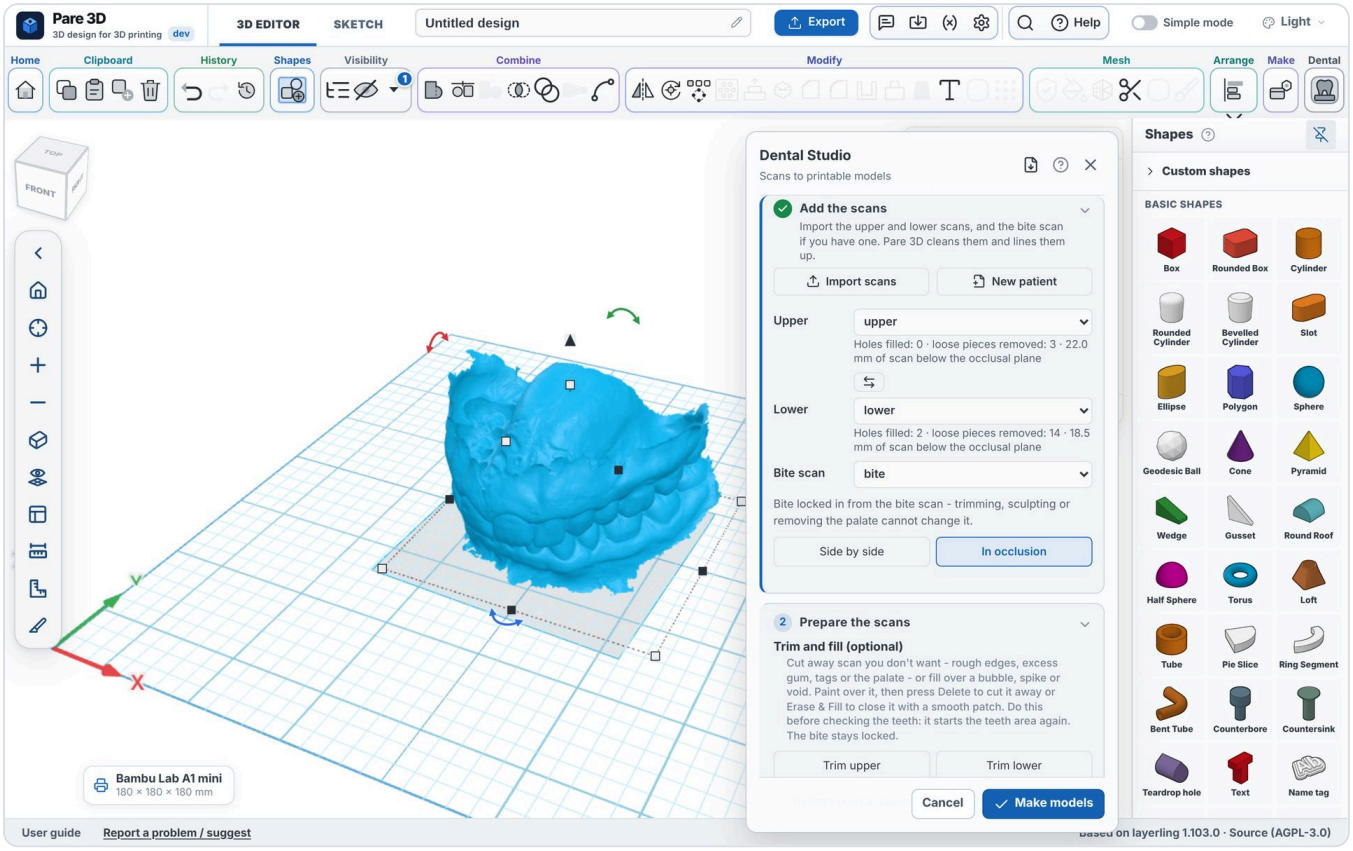

All three slots filled; the bite is locked in.

1. Drag the upper, lower and bite files onto the window together, or click **Import scans** and choose them together. They go straight into the slots.
2. Check the **Upper**, **Lower** and **Bite scan** menus. Each lists every scan in the design and *None*.
3. If upper and lower came out the wrong way round, click the swap button (⇅) between them.
4. Read the line under the slots: *Bite locked in from the bite scan...*, *Upper and lower are in occlusion*, or a warning (chapter 14).

One arch is enough for a single model: leave the other slot on *None*. **New patient** starts a new design with Dental Studio open, so the next patient's scans cannot mix with these.

## Side by side, or in occlusion

Once the bite is found, the scans are laid out **side by side** on the plate - the upper on the left, the lower on the right, each teeth up with its front towards you - and the bite scan is hidden. **In occlusion** puts the pair in the bite to check it; **Side by side** puts them back for trimming. Neither changes the scans.




**In occlusion**: the two scans in the locked bite, to check it from the front and sides.

## How Pare 3D tells the upper from the lower

When you have not chosen the slots yourself, each scan is scored on three cues, strongest first: **which way the teeth point** (scanners place the upper above the lower, so its teeth point down - read on the file's vertical, whether the scanner exported it Z-up or Y-up), **a scanned palate**, and **the width across the canines**. For a pair in occlusion the choice is then checked by the overjet: the upper incisors must stand in front of the lower. Your own choice in the menus always wins.

## Full arch or quadrant

A full arch is a U: its middle is the tongue or palate, with no tooth there. A quadrant or sextant has its teeth at its middle. Pare 3D counts an arch as a quadrant only when at least 2% of its tooth surface lies within 3 mm of the middle of the teeth - a single scrap (a tag of tongue, or what a trim left) is not enough. Before build 75 one point was; on some scans that made every base come out as a quadrant's rectangular block.

## The cleaning report

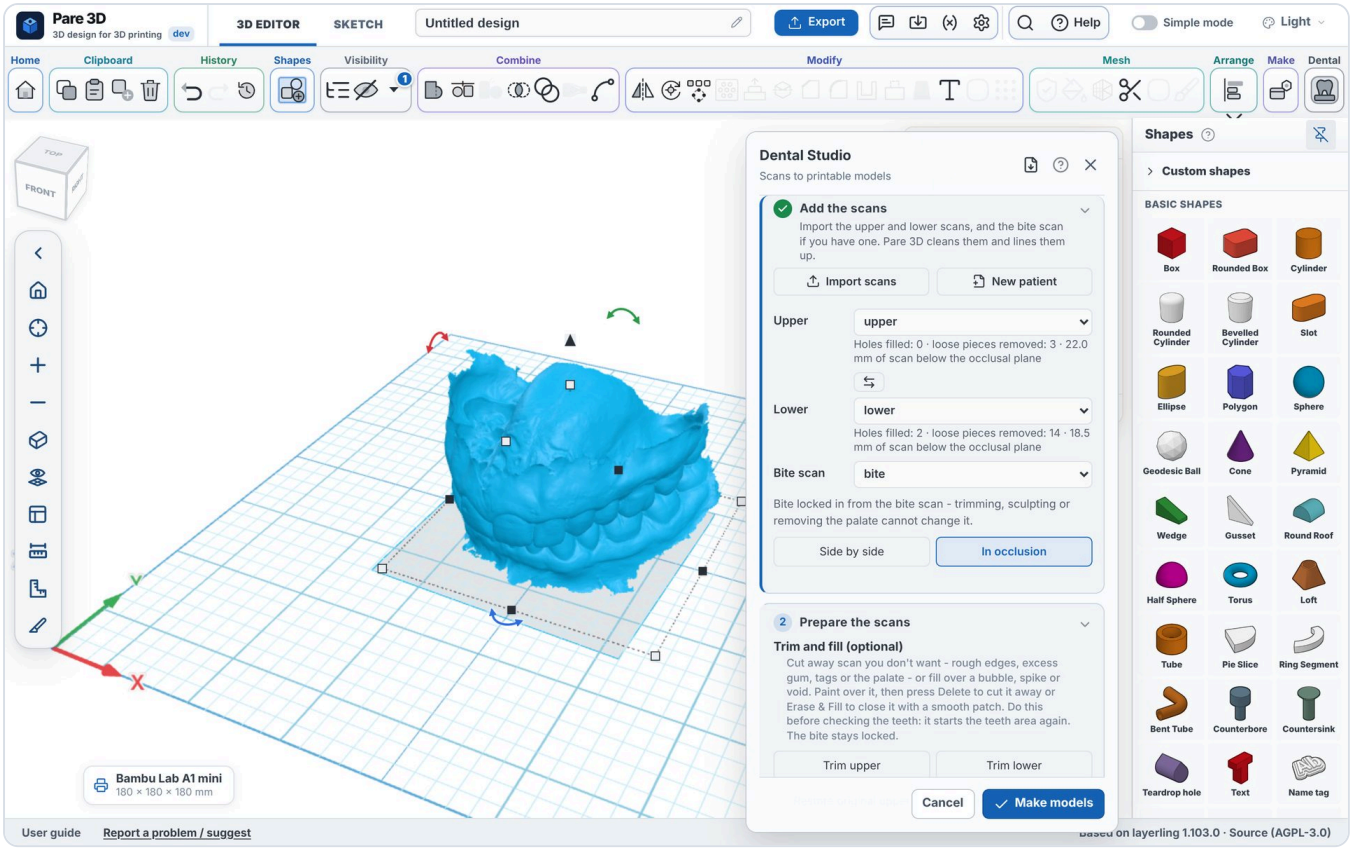
Each arch is cleaned the moment it is chosen. The line under it reads, for example, *Holes filled: 2 · loose pieces removed: 14 · 18.5 mm of scan below the occlusal plane*.

| Item | What was done |
|---|---|
| Holes filled | Every opening in the surface except the scan's outer edge (which becomes the margin side) is closed with new triangles - typically gaps between teeth or in the palate. |
| Loose pieces removed | Islands smaller than 5% of the scan - cheek, tongue, cotton roll and stray captures - are deleted. |
| Broken triangles | Edges shared by more than two triangles (non-manifold geometry) are removed so the model can be closed. |
| mm below the occlusal plane | How deep the scan reaches below the occlusal plane. This limits how much gum can be kept and how tall a base that *follows the scan* can be. |

Each arch is also set in a standard frame: teeth up, occlusal plane level, midline front to back, so the models come out level on the build plate. **Restore original upper** and **Restore original lower** (step 2) put a scan back exactly as it was imported, still in the locked bite.

# 5 The bite: lined up once, locked in

The bite scan is used to put the upper and lower arch into the patient's occlusion, independently of where each arch scan ended up. This is sometimes called "stitching" the bite.




The upper and lower arch scans in the bite they were fitted to (the bite scan itself is hidden).

## The method

Each arch is fitted to the bite scan separately, by **point-to-plane iterative closest point** (ICP) registration, starting from where the scanner left it:

1. About 8,000 points are sampled evenly over the arch.
2. Each point is paired with the nearest point of the bite scan - but **only if both surfaces face the same way** (their normals within about 53°). This stops an upper cusp from pairing with the lower cusp it touches in the bite, which faces the opposite way.
3. The rigid movement (rotation and translation, no scaling) that best closes the distances along the bite surface's normals is solved and applied.
4. This repeats up to 50 times, with the pairing distance narrowing from 2 mm to 1 mm and then 0.5 mm, so the fit starts tolerant and ends precise.

## When the fit is accepted

The fit is used only when **both** arches reach a root-mean-square distance below **0.35 mm** to the bite scan, over enough shared surface. On the author's scans the fit was slightly closer to the bite scan than the scanner's own placement: point-to-plane RMS 0.075 mm against 0.078 mm for the upper and 0.086 mm against 0.094 mm for the lower.

If either arch does not fit, both are left where the scanner placed them and the panel says *The bite scan did not line up with the arches*. Since scanners usually export the arches already in occlusion, the pair is often still correct; check it from the front before making the models.

### Without a bite scan

- If the two arches touch where the scanner put them, Pare 3D treats them as in occlusion (*Upper and lower are in occlusion*).
- If they do not touch, the models are still made, but the bases are not matched to a bite: *The arches do not touch: add the bite scan to put them in occlusion*.
- If the two scans clearly do not share the scanner's coordinates (one was moved or turned in another program), each is made on its own, side by side, without a hinge.

> **Checking the bite yourself**
> Turn the view with the right mouse button (or two fingers on a trackpad) and look from the front and both sides before pressing Make models. The models are made in exactly the relation you see.

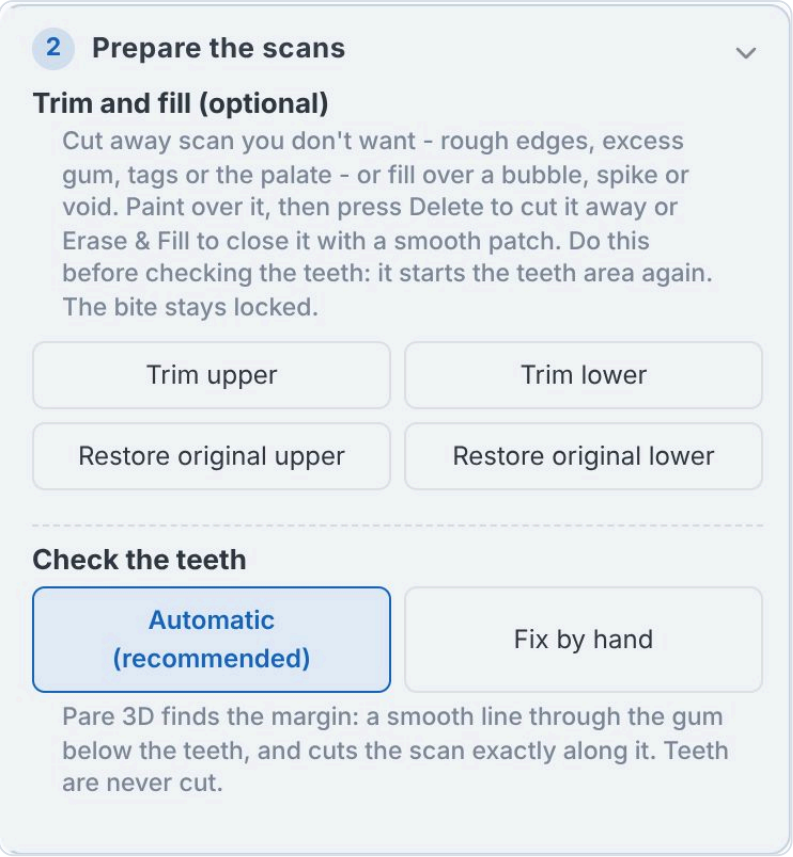
### Locked in

The first time a set is looked at, the bite found is **stored with the two arches**, together with each scan's original file, and is never fitted again. Move the scans apart, turn them, trim rough edges, cut the palate away or sculpt them - the models and the articulator are made in the bite as it was scanned. The panel shows *Bite locked in from the bite scan - trimming, sculpting or removing the palate cannot change it.*

Fitting again after an edit would move it: on a full-arch set, cutting the palate away and fitting again moved the upper by up to 0.33 mm (0.15 mm on average) against the lower. The original files stay in the design (about the size of the STL files, not a copy after every edit), so **Restore original upper** and **Restore original lower** can put a scan back, still in the locked bite.

# 6 Step 2 - Prepare the scans: trim and fill, then check the teeth

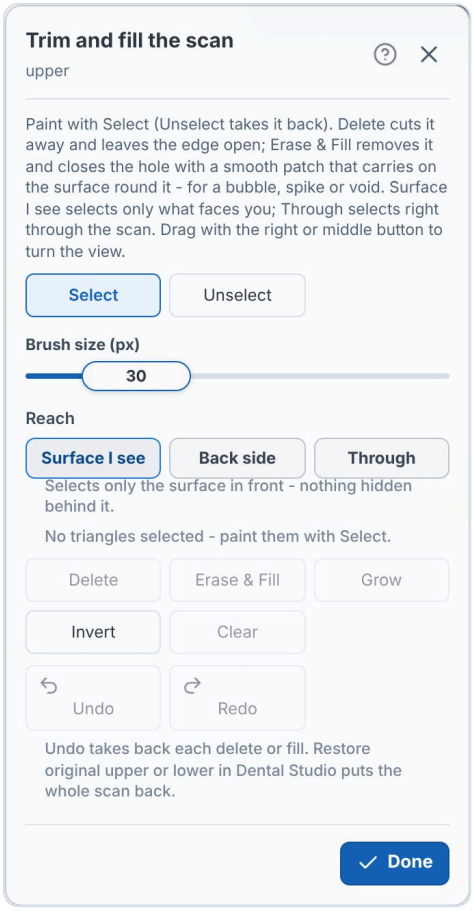
First cut away scan you do not want, or close over a defect; then check what Pare 3D keeps as teeth. In that order: trimming makes the scan anew, so the teeth area is found again after it.




Step 2: Trim and fill (optional) above, Check the teeth below.

## Trim and fill (optional)

**Trim upper** or **Trim lower** opens **Trim and fill the scan** on that arch, with only what this needs. The other scan steps aside while you trim, and the panel stands on the left of the work area, clear of Dental Studio.



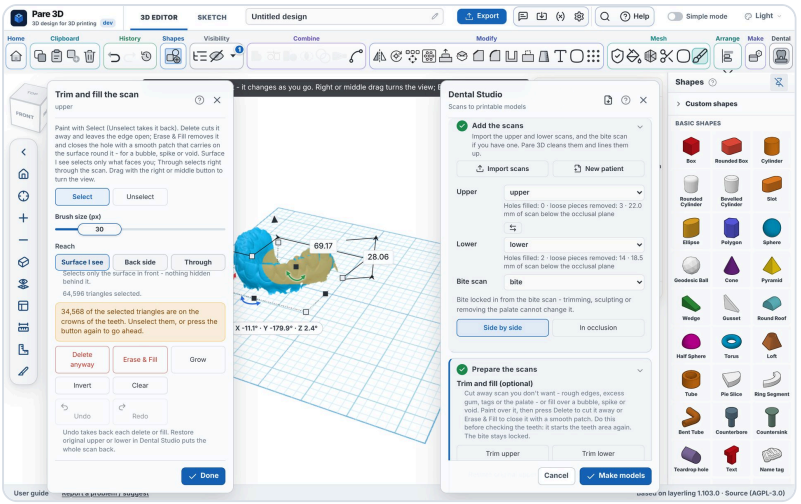

Trim and fill: select, then delete or fill; undo and redo.

| Control | What it does |
|---|---|
| **Select / Unselect** | Paint over what to cut away; Unselect takes it back. |
| **Brush size** | In screen pixels. |
| **Surface I see** | Selects only what faces you - nothing hidden behind it. |
| **Back side** | The side facing away from you, at any depth. |
| **Through** | Every layer under the brush - the quick way to take a palate. |
| **Delete** | Cuts the selection away and leaves the scan's edge open there. |
| **Erase & Fill** | Removes the selection and closes the hole with a smooth patch that carries on the surface round it (Liepa's hole filling, faired as a thin-plate surface[5,6], as Meshmixer does). For a bubble, spike or void. A hole that reaches the scan's own edge is left open. |
| **Undo / Redo** | Take back each delete or fill made since the panel opened, and do it again (`Ctrl` + `Z` / `Ctrl` + `Y` too). They never go back past the panel's opening. |
| **Grow**, **Invert**, **Clear** | Widen the selection by a ring of triangles, swap selected and not, or start again. |

Delete for rough or ragged edges, excess gum, a tag of cheek or tongue, or the palate when the model should be a horseshoe. Erase & Fill for a small defect on the gum or base: a bubble of saliva, a spike, a void where the scanner missed. Keep fills small - a few millimetres; over a large area the patch follows the curve of its rim and can bulge.

**The crowns are guarded**

If what you selected takes in the crowns of the teeth, **Delete** and **Erase & Fill** say how many of the selected triangles lie on them and change nothing. Unselect them, or press the button again (**Delete anyway**) to go ahead. A crown is counted from the cusp tips 4 mm outwards and 6 mm down. Anatomical crowns are about 7.5 mm (molars) to 10-11 mm (incisors and canines) from cusp tip or incisal edge to the cemento-enamel junction, so this stays on the crown; on a short clinical crown it can reach the gingival margin. The gum, the palate and flaps of tissue do not count, nor fewer than 25 triangles at the edge of a stroke. A tall tag of tissue standing right behind the last molar can still be taken for a tooth.




A stroke drawn across the crowns: 34,568 of the 64,596 selected triangles lie on them, and nothing was cut.

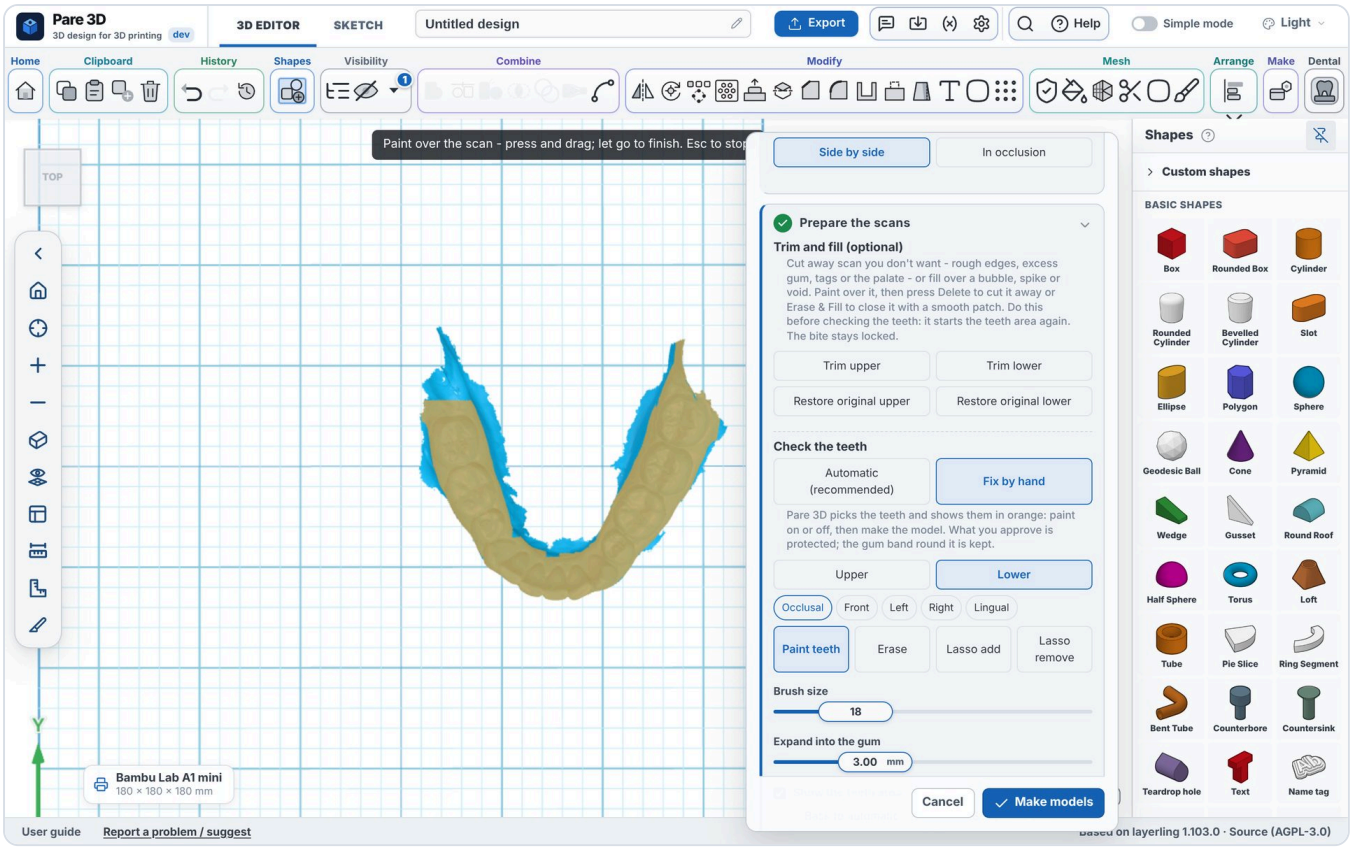
- **The bite stays locked** through every trim (chapter 5).
- **Undo** puts back what was cut or filled; **Restore original upper** / **lower** puts the whole scan back as it was imported.

## Check the teeth - Automatic (recommended)

Pare 3D finds the teeth, then lays a smooth line through the gum below them and cuts the scan exactly along it. Teeth are never cut. How far into the gum the line runs is set in step 4 (**Gum below the teeth**).

## Check the teeth - Fix by hand

Choose **Fix by hand** when the automatic choice missed a tooth surface (a deep gingival recession, a bridge pontic, an implant scan body) or took tissue for tooth (a frenum, a rugae fold). Pare 3D shows its choice as an **orange** tint on the scan.



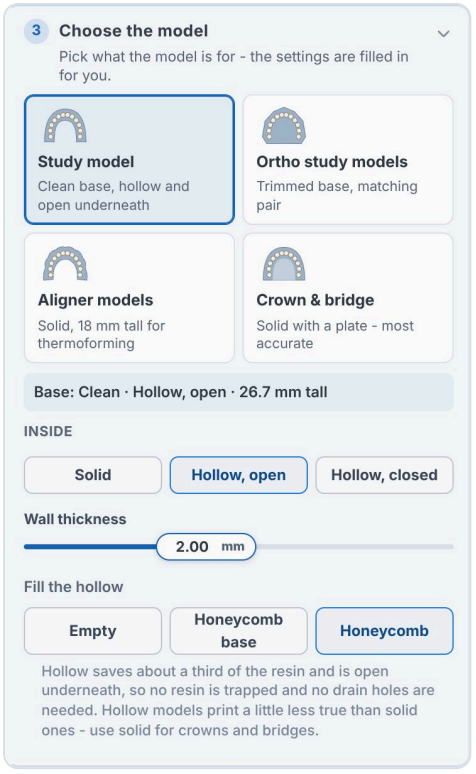

Fix by hand on the lower, from the occlusal side: orange is what the model keeps as teeth.

| Control | What it does |
|---|---|
| **Upper / Lower** | Which arch you are working on. |
| **Occlusal, Front, Left, Right, Lingual** | Turn the view to that side of the chosen arch. |
| **Paint teeth / Erase** | Brush the orange area on or off: drag over the surface. |
| **Lasso add / Lasso remove** | Draw a closed line round an area to add or remove it in one go. |
| **Expand into the gum** | The band of gum kept round the orange area (0.5-10 mm, default 3 mm). The scan is cut smoothly along its outer edge. |
| **Show the teeth area** | Hides the orange tint while you look at the scan; it is never shown while a scan is trimmed. |
| **Back to automatic** | Discards your changes on this arch. |
| **Sculpt this scan** | Opens Sculpt on this arch (chapter 12). |

The brushes change only the surface facing you. Turn the view between strokes with the right or middle mouse button; `Esc` puts the brush down. What you mark as tooth is **protected**: the cut never enters it.

# 7 Step 3 - Choose the model

Pick what the model is for; the base (step 4) and the inside are filled in for that purpose. You can change any of them afterwards.




Each choice shows its base from above; solid or hollow is chosen here too.

| Purpose | Base | Inside | Height |
|---|---|---|---|
| **Study model** | Clean: trimmed to a smooth outline a few mm round the teeth, on a low block | Hollow, open underneath | All anatomy kept, 3 mm base |
| **Ortho study models** | Orthodontic: trimmed the British/ABO way; a pair shares one back line | Solid | Base about a third of the model (6-15 mm) |
| **Aligner models** | Follows the scan | Solid | About 18 mm overall, for thermoforming |
| **Crown & bridge** | With a plate across the arch | Solid | 5 mm base |

Change the base in step 4 and the choice is marked *custom*; pick the model again to go back to its settings.

## Why these defaults

- **Study models** are hollow to save resin: about a third less. They are open underneath, so no resin is trapped and no drain holes are needed.
- **Aligner models** are solid and 15-20 mm tall, as Formlabs recommends for thermoforming: solid models reduce breakage in post-processing, and the height makes the formed appliance easier to remove [1].
- **Crown & bridge** models are solid with a plate across the arch. Hollow models are less true than solid ones, and in hollow models a thicker shell with a cross-arch bar gave the highest accuracy and stability after articulator mounting [2, 3]; solid with a plate is the safest choice where margins and contacts matter.
- **Ortho study models** follow the conventional British/ABO trimming: back square to the midline, heels, buccal sides at 65° (upper) and 55° (lower), the upper's front pointed and the lower's rounded, so the pair stands on its backs in occlusion.

## Inside

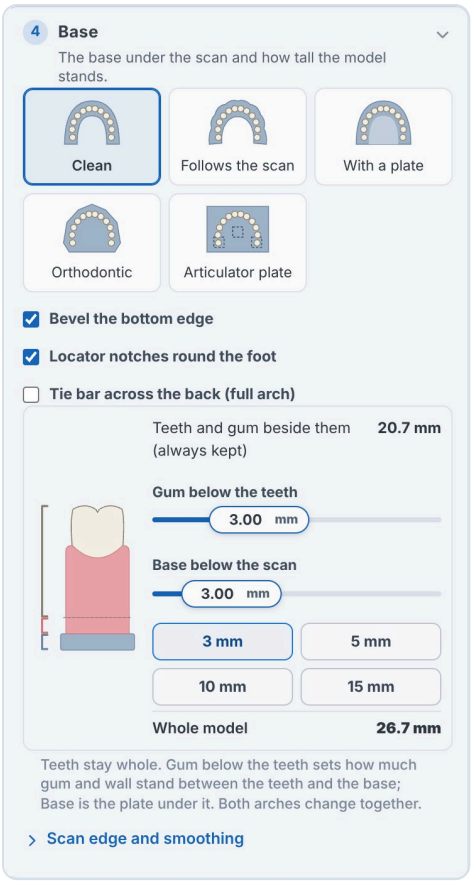
| Choice | What it is |
|---|---|
| **Solid** | Filled throughout. Prints truest. |
| **Hollow, open** | A shell open underneath: about a third less resin, nothing trapped, no drain holes needed. |

| | |
|---|---|
| **Hollow, closed** | A shell with a closed floor and vent holes up into the hollow, so resin drains and air gets in. Needed for a label underneath. |

- **Wall thickness** - 1-5 mm (default 2 mm). Formlabs advises at least 3 mm where a hollow model is thermoformed [1]; thicker shells were more accurate after mounting [2]. The wall is never thinner than set, also round the locator notches.
- **Fill the hollow** - *Empty* saves the most material; *Honeycomb base* stiffens the bottom only; *Honeycomb* carries the whole shell. All drain.

# 8 Step 4 - The base and the model's height

Always open: the base's shape and details, and how tall the model stands.

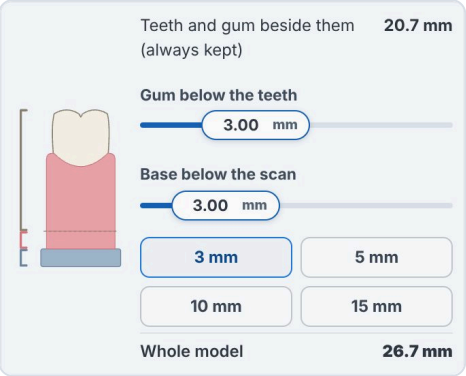



Base styles, details and the height.

| Style | Shape | Best for |
|---|---|---|
| **Clean** | The scan trimmed to a smooth outline a few mm round the teeth, standing on a low block | Study and patient-education models |
| **Follows the scan** | The base wall follows the scan's own edge straight down | Aligners and retainers |
| **With a plate** | A plate across the whole arch, palate or tongue space included | Crown and bridge, mounting - stiffest |
| **Orthodontic** | Trimmed art base (chapter 7) | Ortho records |
| **Articulator plate** | A solid base with pockets underneath for a magnetic articulator plate | Mounting on a physical articulator; always solid |

- **Bevel the bottom edge** - a 0.8 mm bevel round the foot, so a scraper gets under it and the model comes off the build plate without chipping.
- **Locator notches round the foot** - half-round notches front and sides, so a lab can seat the model the same way each time. A hollow model keeps its full wall round them.
- **Tie bar across the back (full arch)** - joins the two back ends at the floor so a full arch cannot spread or warp while it prints. Cut it off afterwards.

## How tall the model is




A section through a molar in its gum on the base, each part bracketed to scale.

- **Teeth and gum beside them** - always kept: the crowns and the gum right beside them, down to where the deepest tooth reaches.
- **Gum below the teeth** - how much more gum and wall stand between the teeth and the base.
- **Base below the scan** - the block or plate under it, with 3, 5, 10 and 15 mm one click away.
- **Whole model** - the three together, from the floor to the tallest tooth. Both arches change together, and a pair stands equally tall.

The height is measured from the teeth, not from the tallest scrap of scan: on the author's set it reads 26.7 mm and the models came out 25.0 and 26.0 mm tall (before build 75 it read 34 mm).

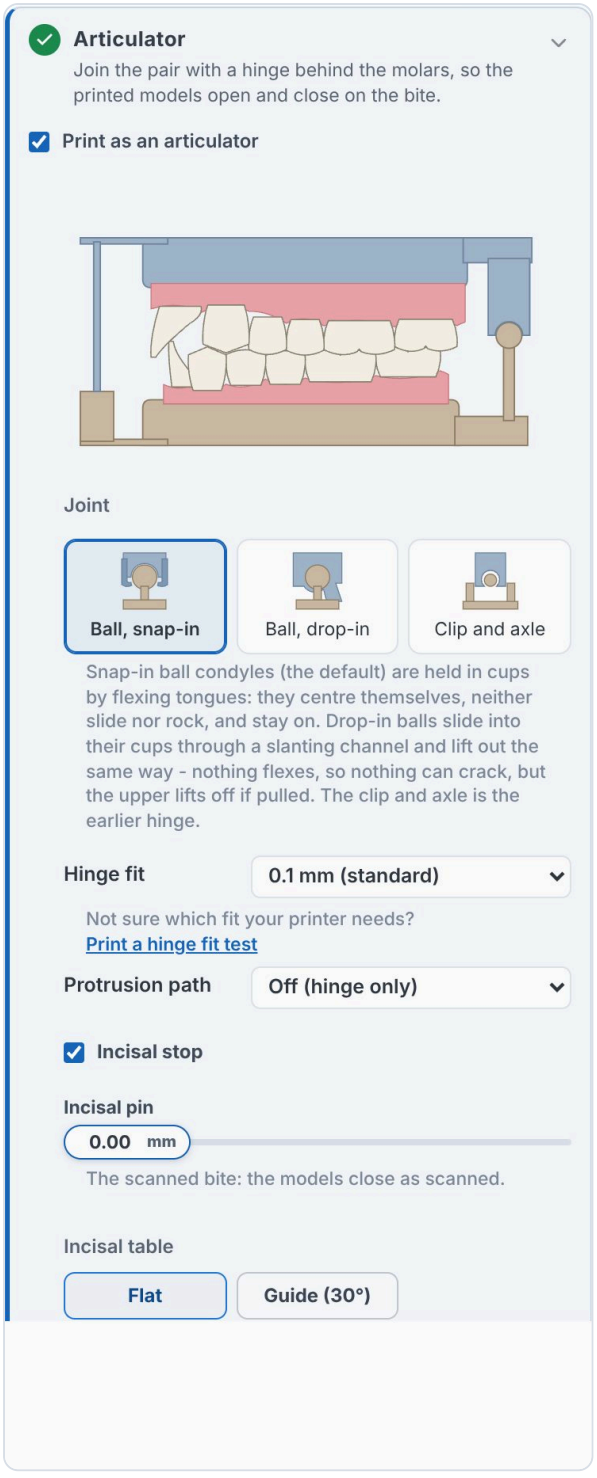
> **Wall too tall?**
> Lower **Gum below the teeth** first, then the base.

## Scan edge and smoothing

- **Follow the gum line (palate and floor of the mouth go)** - a band of gum all round the teeth and nothing more, so the model is a horseshoe; its depth is then the *Gum below the teeth* slider.
- **Smooth the scan's edge** - *Off*, *Low* (default) or *High*: how much the ragged scan border is smoothed where it meets the base. The teeth are never smoothed.

# 9 Step 5 - The articulator

For a pair only. Tick **Print as an articulator** to build a hinge behind the last molars into the two bases, so the printed models open and close on the scanned bite.



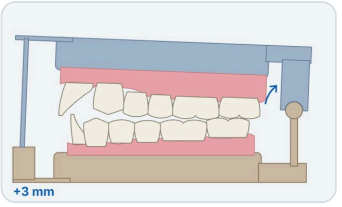

The articulator card: the side view, joint, fit, protrusion, incisal stop and pin, and how the models are placed.

## The side view

The picture is drawn as a lateral view of the pair, front on the left: the incisors in profile with overjet and overbite, canines and molars in a Class I relation along a curve of Spee, the joint behind the last molar and the incisal pin on its table. It follows the settings as you change them.



Pin at 0: the scanned bite.




Pin raised 3 mm: the upper turns up about the joint.

## Joint

| Joint | How it works | Choose it when |
|---|---|---|
| **Ball, snap-in** (default) | Ball condyles held in cups by flexing tongues; self-centring, no slide or rock, stays on | FDM (printed and tested); resin testing under way |
| **Ball, drop-in** | Balls slide into their cups through a slanting channel and lift out the same way; nothing flexes | Brittle resins, where a snap could crack |
| **Clip and axle** | Two posts and an axle on the lower, a clip on the upper | A plain hinge with no protrusion |

**A quadrant against a full arch**

When one arch is a quadrant (or has no teeth on one side) and the other a full arch, the pair is levelled and laid out by the full arch and articulated as a full pair: a condyle behind each side of the full arch, the incisal pin on its midline. The quadrant's base grows into a horseshoe 8 mm wide that follows the full arch's base (tongue space left open) and carries both condyle posts and the incisal table. Hinge, hinge and pin hold the models in the scanned bite, so they cannot rock or flex on the side with no opposing teeth; the incisal stop is always on for such a pair.

## Fit, movements and stops

| Option | Values | What it does |
|---|---|---|
| **Hinge fit** | 0.05-0.3 mm (0.10 standard) | The clearance round the joint. 0.10 mm turns without play on a well-tuned printer; too stiff, choose a looser fit and make the models again. **Print a hinge fit test** adds a strip with five fits to find yours. |
| **Protrusion path** | Off (hinge only), 10-45° in 5° steps | Lets the lower slide forward along a condylar path at this angle below the occlusal plane, to check protrusive contacts. Protrusive records average about 30° [4]. |
| **Incisal stop** | On / off | A slim pin in front of the models meets a table as they close, so the pair closes to the scanned bite every time and the pin, not the teeth, takes the load. |
| **Incisal pin** | 0 (scanned bite) to 10 mm, in 0.5 mm steps | Opens the bite at the pin (a raised vertical dimension, for a splint or a rehabilitation). As on an articulator, the bite opens by turning about the joint: the front teeth part most and the molars least (roughly pin : front teeth : back molars = 3 : 2 : 1). After making, the notice gives the exact figures. |
| **Incisal table** | Flat, Guide (30°) | A guide table slopes up towards the back, so the models part as the lower slides forward, as the anterior guidance does. |

## Place the models

- **Ready to print** - upper and lower side by side, teeth up, flat on the build plate.
- **In occlusion** - stacked in the bite, as in the mouth; with a raised pin the upper stands open on it. The default for ortho models.

After making, the notice says how far the articulator opens, e.g. *The articulator opens to 90° or more*; under 30° before the models meet, Pare 3D warns you to check the hinge.

**Printed results**

The snap-in ball joint has been printed successfully on an FDM printer. On resin, start with the fit test, and choose the drop-in ball for brittle resins.

# 10 Step 6 - Name the model

A label on the base, or no patient details at all.



Patient details and the label.

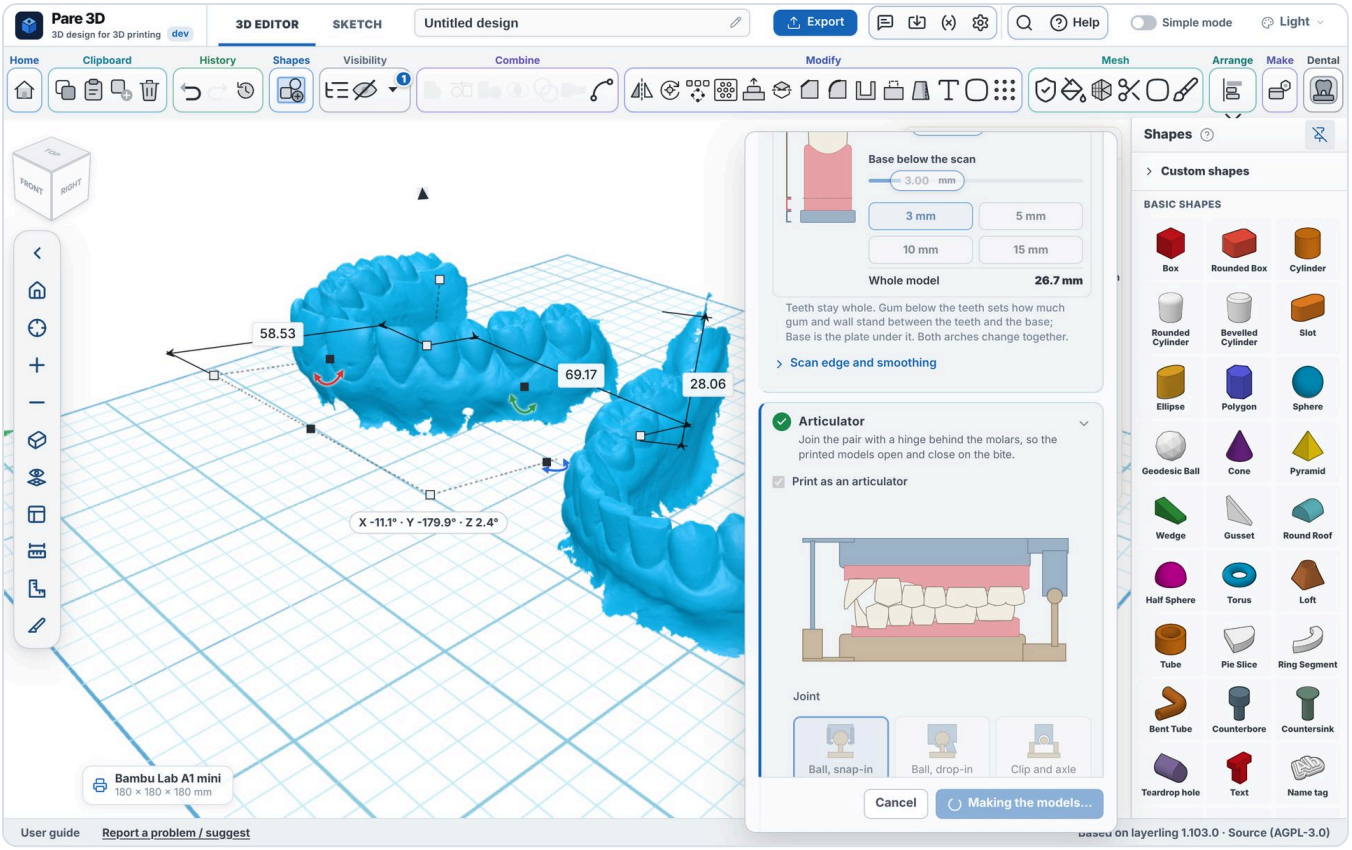
- **Patient details: Keep** names the models after the scan files (often the patient's name) and engraves the label.
- **Patient details: Remove** names them *Upper model* and *Lower model*, renames the bite scan and engraves no label, so the models and every file you export carry nothing that identifies the patient. Pare 3D remembers the choice.
- **Label** - a patient code or a date, engraved on the front of the orthodontic and articulator bases (with U or L in front for a pair) or under the floor on the others. A hollow model open underneath has nowhere to carry it, so with a label its floor is closed, with vent holes.

**One patient per design**

Keep each patient's scans in a design of their own: **New patient** in step 1 starts one with Dental Studio open. Give the design a case number rather than a name.

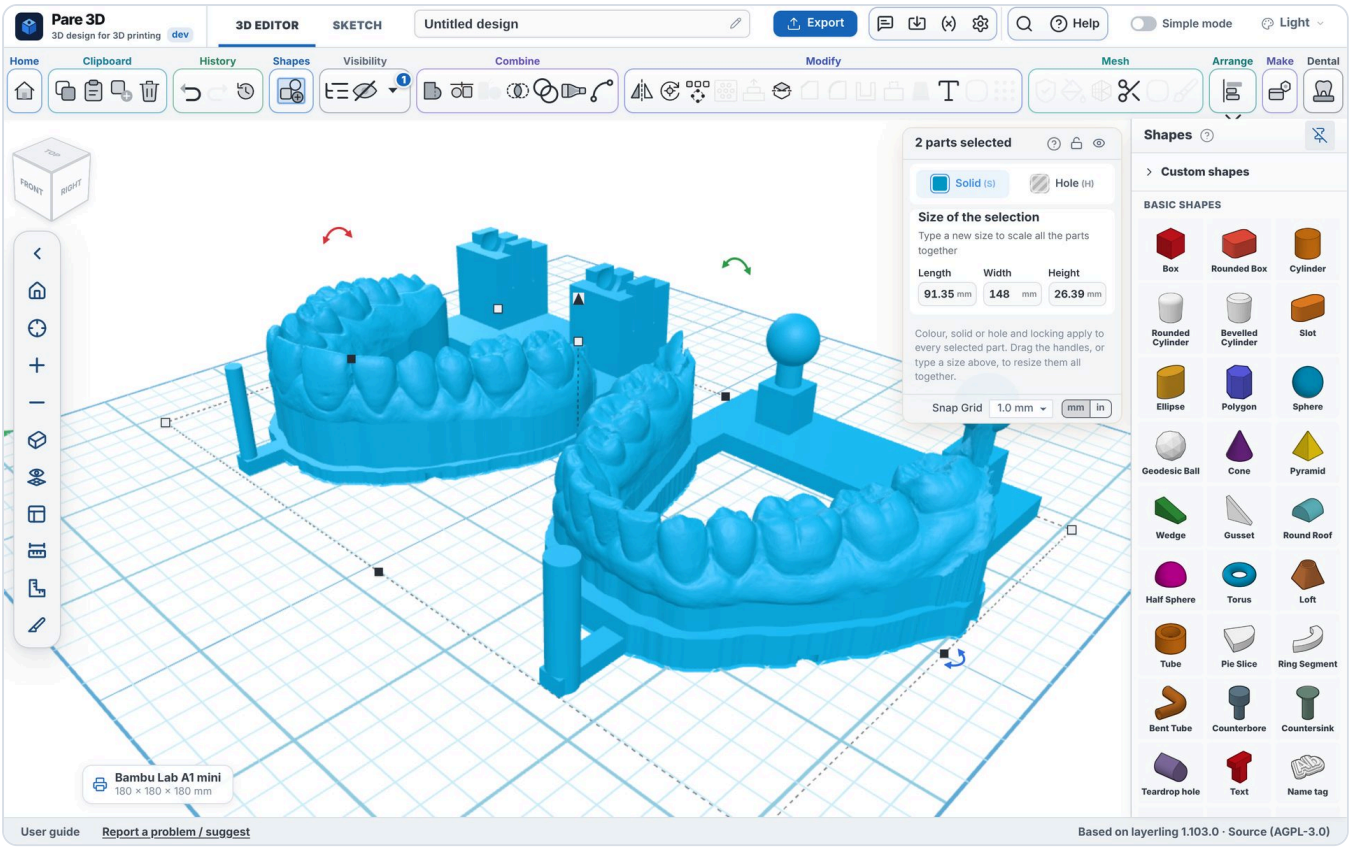
# 11 Making the models

Press **Make models** (or **Make model** for one arch). The button stays at the bottom of the panel, always on screen.




While the models are made you can still turn the view.

The two arches are made at the same time in two background workers. Then the bases are matched, the articulator parts added, and the models placed. The panel closes; the new models are selected and the scans they came from are replaced (the bite scan is hidden, not deleted). Before a model is handed over it is checked against its scan and written as one clean closed solid; flecks under 5 mm³, sealed bubbles and small tags of scan standing free of the model are left out. Where the gum cut passes within a fifth of an edge of a corner, the corner is moved onto the line rather than a sliver split off, so the margin is made of well-shaped triangles.




The finished pair, ready to print: hollow study bases, snap-in ball articulator, incisal pin and table.

- **Undo** (`Ctrl` / `Cmd` + `Z`) brings the scans back exactly as they were; change a setting and make the models again.
- To change a finished model, select its scan again: Dental Studio works from scans, not from finished models.
- The size of the selection is shown in the side panel; typing a size there scales the models - do not, if they must stay true to the patient.

# 12 Editing a scan by hand: Sculpt

Sculpt reshapes areas of a scan before the models are made - a spike, a fold of cheek, a rough area. To cut scan away, Trim (chapter 6) is simpler.

Open it from **Sculpt this scan** in Fix by hand, the brush button in the **Mesh** toolbar group, or the right-click menu. The panel opens at once and says *Getting the mesh ready to sculpt…* while a large scan is prepared (about a second for a full arch).

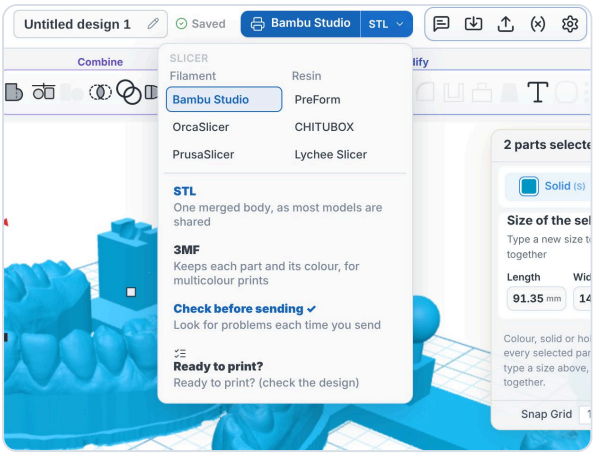
| Brush | Effect |
|---|---|
| Draw, Inflate, Deflate | Raise the surface, or push it out or in along its normals. |
| Smooth | Evens out bumps and spikes up to the brush's size; at full strength removes fissure detail. |
| Flatten, Pinch, Crease | Pull the surface onto a plane, together, or into a sharp line. |
| Move | Drag a patch of surface with the brush. |
| Select / Unselect | Paint triangles to delete, erase and fill, smooth or flatten as a group. |

- **Refine** adds triangles where the brush works, so fine detail can be shaped on a coarse part; **Symmetry** mirrors strokes left-right or front-back.
- Strokes show at once on the full-detail scan. **Undo** and **Redo** (or `Ctrl` / `Cmd` + `Z` and `Y`) step through strokes one at a time, then the deletes and fills made since the panel opened; the scan is rebuilt once when you press **Done**.
- Sculpting one of the Studio's scans hides the other until Sculpt closes; the bite stays locked.

> **Never sculpt the teeth you need to be true**
> Sculpt is for artefacts and soft tissue. Smoothing or flattening a tooth changes the anatomy the model reproduces.

# 13 Printing

Send the models straight to your slicer from the title bar.




The send menu: slicer and file type.

1. Open the small menu beside the Send button and choose the slicer: **Bambu Studio, OrcaSlicer, PrusaSlicer** (filament) or **PreForm, CHITUBOX, Lychee Slicer** (resin). A slicer not found on this computer is marked *not found*.
2. Choose **STL** (one merged body) or **3MF** (each part separately; the resin slicers read 3MF too).
3. Click **Send to ...**. The app saves a copy in *Documents › Pare 3D Exports* and opens it in the slicer.

With **Check before sending** on, *Ready to print?* first looks for floating parts, thin walls, large overhangs and parts past the build plate.

In a web browser the file is downloaded instead; open it in the slicer yourself.

## Printing tips

| | Resin (SLA / DLP / LCD) | Filament (FDM) |
|---|---|---|
| Orientation | Flat on the plate (Ready to print) for hollow open models; resin drains from underneath. Solid models can be printed flat with no supports. | Flat, teeth up. |
| Hollow | Use hollow open, or hollow closed with vents; never a closed shell without vents. | Solid or a honeycomb is usually faster than shells. |
| Articulator | Print a hinge fit test first; drop-in ball for brittle resins. | Snap-in ball, 0.10 mm, has printed successfully. |
| Removal | Keep Bevel the bottom edge on. | - |

# 14 Messages and what to do

| Message | Meaning and what to do |
|---|---|
| The bite scan did not line up with the arches | One arch fitted the bite worse than 0.35 mm RMS, or they share too little surface. The scanner's positions are used. Check the bite from the front; re-export the bite scan from the same case if it is wrong. |
| The two scans are not where the scanner put them... | The arches do not share coordinates (one was moved or turned). Each is made on its own, without bite or hinge. Re-import the original exports. |
| The arches do not touch: add the bite scan... | No bite scan and the arches are apart. Add the bite scan in step 1. |
| One scan must be the upper and the other the lower | Both slots name the same arch. Use the swap button or the menus. |
| Part of the scan's ragged edge could not stand on the base... | The scan was cut where its edge is complete. Check that nothing you need was lost; rescan deeper if it was. |
| Some of the tooth surface could not be kept where the scan's edge folds under the teeth | An undercut or a fold at the edge. Check the model; trim the fold (step 2, Trim) and make it again. |
| There was no room for vent holes | A hollow closed model too small for vents. Use hollow open or solid. |
| The label was left off... | Labels need the orthodontic base, or a plate with a solid or closed inside, and room for letters at least 1.5 mm tall. |
| The articulator opens less than 30° before the models meet | Something blocks the opening (tall distal tissue, a tie bar). Check the hinge from the side before printing. |
| Part of a model came out loose from its base (a piece of the hinge or the stop) | With the articulator on: a piece of the hinge or the stop is not joined. Do not print; report it with the scans (Report a problem in the footer). |
| Part of a model came out loose from it - a piece of scan standing free of the model | Without an articulator: a piece of scan too big to be a mere tag stands free. Trim the scan there (step 2) and make the model again. |
| A small tag of scan standing free beside a model was left out | A sliver of cheek or tongue at the end of an arch, walled down to the floor on its own (under 150 $mm^3$ and a twentieth of the model), held no teeth and was left out. Nothing to do. |
| ... of the selected triangles are on the crowns of the teeth | Trim stopped before cutting or filling over crowns. Unselect them, or press the button again to go ahead. |
| Erased ... triangles and filled ... hole(s); ... opening(s) ran into the scan's edge and were left open | Erase & Fill closed what it could; a hole that reaches the scan's own edge has nothing to close against. Use Delete there instead. |
| A large mesh - It has ... triangles; cutting, grouping and merging handle ... | The scan is over the triangle limit for cutting and grouping. Raise it (chapter 3); Dental Studio works either way. |
| The scans changed while the models were being made | A scan was edited during making. Make the models again. |

# 15 Quick reference

**The fastest route**

1. Home › Dental models from scans
2. Drag the three scan files onto the window
3. Check: *Bite locked in from the bite scan*
4. Trim or fill if needed (step 2)
5. Pick the purpose
6. Tick Print as an articulator (pairs)
7. Make models
8. Send to your slicer

**Defaults by purpose**

| | Base | Inside |
|---|---|---|
| Study | Clean, 3 mm | Hollow, open |
| Ortho | Orthodontic | Solid |
| Aligner | Follows scan, ~18 mm | Solid |
| C&B | Plate, 5 mm | Solid |

**The six steps**

1 Add the scans · 2 Prepare (trim and fill, then teeth) · 3 Choose the model (and inside) · 4 Base and height · 5 Articulator (pairs) · 6 Name the model. Click a title to fold it.

**Bite**

Point-to-plane ICP per arch, normals must agree, pairing radius 2 → 1 → 0.5 mm, up to 50 rounds, accepted below 0.35 mm RMS. Fitted once, then locked in with the original scans.

**Articulator settings**

Joint: snap-in ball (default), drop-in ball, clip and axle. Fit 0.05-0.3 mm (0.10 standard). Protrusion off or 10-45°. Incisal stop; pin 0-10 mm in 0.5 mm steps, opening about the joint; table flat or 30° guide.

**Keys**

`Ctrl` + `K` search · `Ctrl` + `Z` undo (in Trim and Sculpt: back to the panel's opening) · `Esc` put a brush down · right or middle drag: turn the view · wheel: zoom.

# 16 References


1. Formlabs. Thermoformed clear aligners and retainers - application guide. *dental.formlabs.com/indications/thermoformed-clear-aligners-retainers/guide* (accessed 26 October 2026). Models 15-20 mm tall; solid models recommended; hollow walls at least 3 mm.
2. Wan Q, Riera C, Tran TT, Fehmer V, Sailer I, Lee JH. Printing accuracy and post-mounting dimensional stability of 3D-printed hollow dental models with varying shell thicknesses and cross-arch bars. *J Dent*. 2026;164:106234. doi:10.1016/j.jdent.2025.106234
3. Yurdakurban E, Topsakal KG, Duran GS, Görgülü S. Evaluating the accuracy between hollow and solid dental aligner models: a comparative study of printing technologies. *Aust Orthod J*. 2024;40(2):51-62. doi:10.2478/aoj-2024-0023
4. Kwon OK, Yang SW, Kim JH. Correlation between sagittal condylar guidance angles obtained using radiographic and protrusive occlusal record methods. *J Adv Prosthodont*. 2017;9(4):302-307. doi:10.4047/jap.2017.9.4.302 (protrusive records: 30.1° ± 6.7° right, 30.2° ± 6.3° left).
5. Liepa P. Filling holes in meshes. In: *Proceedings of the Eurographics/ACM SIGGRAPH Symposium on Geometry Processing*. Aachen: Eurographics Association; 2003:200-206. doi:10.2312/SGP/SGP03/200-206
6. Kobbelt L, Campagna S, Vorsatz J, Seidel HP. Interactive multi-resolution modeling on arbitrary meshes. In: *Proceedings of SIGGRAPH 98*. New York: ACM; 1998:105-114. doi:10.1145/280814.280831


The registration, cleaning and margin methods described in chapters 4-6 are those implemented in Pare 3D build 77 (files *dentalArch* and *dentalModel* in the source code, AGPL-3.0). Pare 3D is not a medical device; check every model before clinical use.